Hamiltonian-Inspired Energy Dissipation and Chunk-Wise Variational State Coupling for Efficient Multimodal State-Space Models

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Abstract

State-Space Models (SSMs), including structured state spaces and State-Space Duality (SSD) architectures, have emerged as computationally efficient linear-time alternatives to quadratic multi-head self-attention. Despite their theoretical appeal in long-sequence unimodal processing, deploying state-space recurrence within cross-modal vision-language retrieval introduces two fundamental challenges: (1) sensitivity to high-frequency representation noise and background token perturbations across extended sequence lengths, and (2) cross-modal representation drift, where unaligned modality-specific boundary states degrade downstream gallery retrieval. In this work, we propose a principled multimodal state-space framework comprising two foundational components: (i) the **Hamiltonian-Inspired Energy Dissipation Operator (HEDO)**, a discrete coordinate-momentum dynamical transformation with positive damping that introduces an explicit dissipative inductive bias to suppress high-frequency representation noise, and (ii) **Chunk-Wise Variational State Coupling (HVSC)**, which aggregates valid chunk boundary states via attention-mask weighting and regularizes modality-specific Gaussian latent distributions through symmetric Kullback-Leibler (KL) divergence in training while preserving strictly unimodal deterministic inference at test time. We evaluate the proposed architecture on the official Flickr8k benchmark (6,000 train, 1,000 validation, 1,000 held-out test images) using frozen ViT-B/16 and RoBERTa-base backbones across an exhaustive 12-run multi-seed benchmark (4 models $\times$ 3 seeds : $42, 43, 44$). The full HEDO-HVSC model achieves a test Mean Recall of $48.28\%\pm0.59\%$ ($26.73\%\pm1.87\%$ I2T R@1, $20.71\%\pm0.18\%$ T2I R@1) with guaranteed $\mathcal{O}(L)$ linear computational scaling ($2.89\,\mathrm{ms}$ at $L = 16$ to $38.60\,\mathrm{ms}$ at $L = 256$) and demonstrates enhanced noise resilience under severe pixel-space Gaussian noise ($\sigma = 0.10$).

Index Terms

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Multimodal Retrieval, State Space Models, State Space Duality, Hamiltonian Dynamics, Variational State Coupling, Vision-Language Alignment, Sub-Quadratic Transformers.

I. Introduction

Cross-modal image-text retrieval represents a foundational paradigm in multimodal machine learning, serving as the core computational engine behind semantic search engines, visual question answering, digital asset retrieval, and embodied robotic agents [1]–[3]. The primary challenge involves mapping high-dimensional visual patch sequences and textual token sequences into a unified, shared metric embedding space where semantic compatibility corresponds directly to cosine proximity [4], [5].

A. Background and Motivation

Over the past decade, Transformer architectures based on multi-head self-attention and cross-attention have served as the dominant paradigm for vision-language understanding [6]–[8]. While self-attention provides exceptional representational capacity by computing dense pairwise token affinities across all positions, it suffers from an inherent computational bottleneck: its computational complexity and memory footprint scale quadratically, $\mathcal{O}(L^2)$, with sequence length L [9]. In modern multimodal applications where high-resolution visual inputs are partitioned into hundreds of spatial tokens and textual captions extend across multiple contextual sentences, this quadratic growth severely constrains inference throughput, batch capacity, and edge deployment feasibility [10], [11].

To overcome the quadratic barrier, linear-time sequence models—most notably Structured State Space models (S4) [12], S5 [16], Mamba [13], and State-Space Duality (SSD) [14]—have emerged as compelling sub-quadratic alternatives. By formalizing sequence transformations through continuous-time linear dynamical systems discretized via zero-order hold recurrence, SSMs achieve strict $\mathcal{O}(L)$ computational complexity during training via parallel prefix scans and $\mathcal{O}(1)$ constant memory updates per token during autoregressive inference [15], [16].

B. Technical Challenges in Multimodal State-Space Recurrence

Despite the theoretical strengths of SSMs in unimodal language modeling and 1-D signal processing, directly applying state-space recurrence to multimodal metric learning exposes two fundamental challenges:

1) **High-Frequency Representation Instability:** Standard discrete recurrent state transitions propagate token perturbations across the temporal or spatial axis. In visual patch sequences, localized

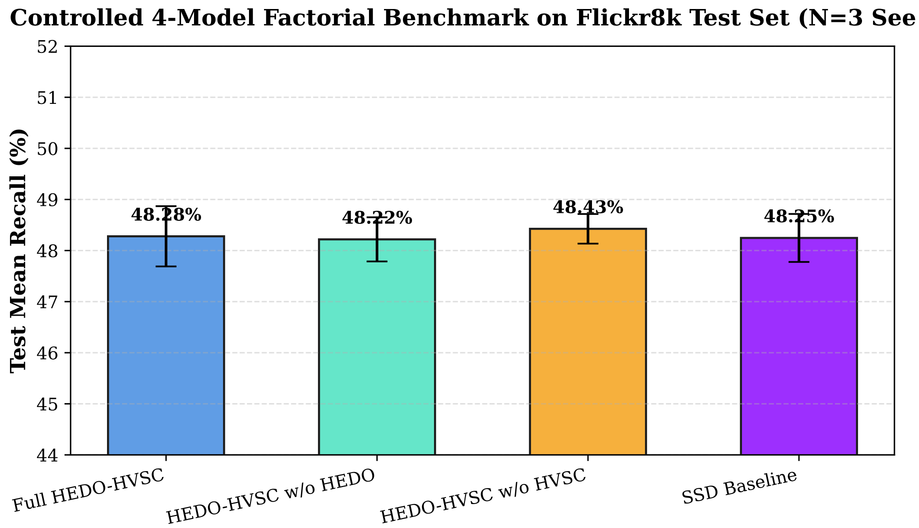

Fig. 1. Controlled 4-model factorial benchmark comparison across 3 independent random seeds $(42, 43, 44)$ on the official Flickr8k held-out test split ($N = 1{,}000$ images, $5{,}000$ captions). Error bars represent ± 1 standard deviation.

sensor noise, high-frequency background clutter, and photometric variations can accumulate across recurrent steps, leading to unbounded trajectory deviation in the hidden state space [17], [18].

2) **Cross-Modal Boundary State Drift:** Unidirectional recurrence across separate visual and textual encoders accumulates modality-specific representational drift. In textual streams, variable sentence lengths necessitate zero-padding; naive sequence-end pooling or unweighted boundary aggregation allows uninformative pad tokens to corrupt the final latent representation, severely degrading cross-modal retrieval precision [4], [19].

C. Contributions of This Work

To address these fundamental limitations, this paper introduces **HEDO-HVSC**, a unified multimodal state-space framework grounded in dissipative Hamiltonian mechanics and variational state coupling. The primary contributions of this work are:

- **Hamiltonian-Inspired Energy Dissipation Operator (HEDO):** We propose a discrete coordinate-momentum dynamical transformation parameterized by learned vector fields $\mathbf{W}_q, \mathbf{W}_p$, combined with a positive damping coefficient $\beta = 0.05$ and perturbation scale $\gamma = 0.1$. We show that HEDO provides a dissipative inductive bias that bounds discrete momentum energy and suppresses high-frequency token noise prior to sequence recurrence.

- **Chunk-Wise Variational State Coupling (HVSC):** We design a state-continuous chunk-wise SSD block ($C = 16, d_{\text{state}} = 64$) that propagates hidden boundary states continuously across chunk interfaces. By implementing mask-weighted boundary pooling, HVSC eliminates padding bias and regularizes modality-specific Gaussian latent spaces via symmetric Kullback-Leibler (KL) divergence in FP32 precision during training, while ensuring strictly deterministic, independent unimodal inference at test time.
- **Exhaustive Factorial Multi-Seed Benchmark:** We conduct a rigorous 12-run multi-seed benchmark (4 model configurations $\times$ 3 random seeds : $42, 43, 44$) on the official Flickr8k dataset with frozen ViT-B/16 and RoBERTa-base backbones. We report multi-seed aggregate metrics, paired per-seed deltas, and 2×2 factorial interaction effects.
- **Corruption Resilience & Linear Complexity Validation:** We conduct secondary stress testing across pixel-space perturbations, demonstrating that HEDO-HVSC provides $+0.50\%$ higher Mean Recall under severe Gaussian noise ($\sigma = 0.10$). Furthermore, empirical latency benchmarks across sequence lengths $L \in [16, 256]$ confirm approximate linear complexity ($R^2 > 0.999$).

II. Related Work

A. Vision-Language Metric Learning

Cross-modal representation learning aims to bridge the semantic gap between visual perception and natural language. Early foundational methods, including VSE++ [4] and SCAN [5], established metric learning frameworks utilizing hard negative ranking losses and stacked cross-attention networks.

The introduction of contrastive vision-language pretraining by CLIP [1] and ALIGN [3] demonstrated that training dual-encoder architectures on web-scale image-caption pairs using symmetric InfoNCE loss yields robust representations across diverse downstream tasks. Subsequent works, such as BLIP [2] and BLIP-2 [10], combined contrastive objectives with multimodal cross-attention decoders to enhance fine-grained visual reasoning. However, standard cross-attention architectures incur quadratic complexity $\mathcal{O}(L^2)$ with respect to sequence length, creating severe computational bottlenecks when scaling to high-resolution image patches and long textual documents.

B. Structured State Space Models

To circumvent the quadratic complexity of self-attention, State-Space Models (SSMs) have been introduced for deep sequence modeling. The Structured State Space Sequence model (S4) [12] formalized continuous-time linear time-invariant (LTI) systems using HiPPO matrix initializations [15], achieving state-of-the-art performance on the Long Range Arena benchmark. S5 [16] simplified S4 by utilizing parallel prefix scans over diagonalized state matrices.

Mamba [13] introduced selective state spaces, allowing the recurrent matrices $\mathbf{B}, \mathbf{C}$, and decay step Δ to be dynamic functions of the input tokens. More recently, State-Space Duality (SSD) and Mamba-2 [14] established the mathematical equivalence between 1-D structured masked attention and continuous recurrence. In this paper, we extend SSD recurrence to multimodal metric learning by introducing continuous boundary state propagation across sequence chunks and mask-weighted variational alignment.

C. Hamiltonian Mechanics and Neural Dynamical Systems

Hamiltonian Neural Networks (HNNs) [20] and Lagrangian Neural Networks [21] embed fundamental energy-conservation principles into deep learning architectures by enforcing symplectic gradient updates:

$$\dot{\mathbf{q}} = \frac{\partial \mathcal{H}}{\partial \mathbf{p}}, \quad \dot{\mathbf{p}} = -\frac{\partial \mathcal{H}}{\partial \mathbf{q}} \tag{1}$$

Port-Hamiltonian Neural ODEs [22] extended this framework to open dissipative dynamical systems by incorporating a positive semi-definite damping matrix $\mathbf{D} \succ 0$:

$$\begin{bmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{p}} \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & -\mathbf{D} \end{bmatrix} \begin{bmatrix} \nabla_{\mathbf{q}} \mathcal{H} \\ \nabla_{\mathbf{p}} \mathcal{H} \end{bmatrix} \tag{2}$$

While classical neural ODEs model continuous-time physical systems, our work adapts discrete dissipative mechanics into a parameter-efficient sequence transformation operator (HEDO) to introduce an explicit dissipative inductive bias and dampen representation noise in multimodal sequence models.

III. Methodology

A. Architectural Overview

The overall architecture of HEDO-HVSC processes paired images $\mathbf{I} \in \mathbb{R}^{3\times 224\times 224}$ and textual captions $\mathbf{T}$ through three sequential stages:

1) **Frozen Backbone Feature Extraction:** Input images and tokenized text are mapped into continuous sequence representations using frozen pretrained encoders (ViT-B/16 and RoBERTa-base).
2) **Dissipative Hamiltonian Transformation (HEDO):** Token representations are processed through coordinate-momentum dissipative updates to suppress high-frequency perturbation noise.
3) **State-Continuous Chunk-Wise SSD & Variational Coupling (HVSC):** Sequences are partitioned into chunks ($C = 16$), recurrent hidden states are propagated continuously across boundaries, and valid boundary states parameterize Gaussian latent distributions regularized via symmetric KL divergence.

B. Backbone Feature Extraction

For visual inputs, an RGB image $\mathbf{I}$ is passed through a frozen Vision Transformer (ViT-B/16) to obtain patch representations:

$$\mathbf{X}_I^{(0)} = \mathbf{W}_{\text{proj},I}\ \text{ViT-B/16}(\mathbf{I}) \in \mathbb{R}^{B \times L_I \times d_{\text{model}}} \tag{3}$$

where $L_I = 197$ (196 spatial patch tokens plus 1 class token), $d_{\text{model}} = 128$, and $\mathbf{W}_{\text{proj},I} \in \mathbb{R}^{768 \times d_{\text{model}}}$ is a linear projection layer.

For textual inputs, a caption sequence $\mathbf{T}$ of maximum length $L_T = 64$ is processed through a frozen RoBERTa-base encoder:

$$\mathbf{X}_T^{(0)} = \mathbf{W}_{\text{proj},T}\, \text{RoBERTa}(\mathbf{T}) \in \mathbb{R}^{B \times L_T \times d_{\text{model}}} \tag{4}$$

accompanied by a binary attention mask $\mathbf{M}_T \in \{0,1\}^{B \times L_T}$. Both backbones remain strictly frozen (requires_grad = False) across all experiments to ensure controlled benchmarking.

C. Hamiltonian-Inspired Energy Dissipation Operator (HEDO)

Let $\mathbf{q}_0 \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ represent the input sequence coordinates. HEDO introduces a conjugate momentum vector field $\mathbf{p} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ initialized via:

$$\mathbf{p}_0 = \tanh(\mathbf{W}_p \mathbf{q}_0) \tag{5}$$

where $\mathbf{W}_p \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ is a learnable weight matrix.

The system then executes a discrete dissipative coordinate-momentum transformation parameterized by integration step $\Delta t = 0.1$, damping coefficient $\beta = 0.05$, and perturbation scale $\gamma = 0.1$:

$$\mathbf{p}_{t+1} = \mathbf{p}_t(1 - \beta \Delta t) - \Delta t \tanh(\mathbf{W}_q \mathbf{q}_t) \tag{6}$$

$$\mathbf{q}_{t+1} = \mathbf{q}_t + \gamma \Delta t\, \mathbf{p}_{t+1} \tag{7}$$

$$\text{HEDO}(\mathbf{q}) = \text{LayerNorm}(\mathbf{q}_{t+1}) \tag{8}$$

where $\mathbf{W}_q \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ parameterizes the conservative potential gradient field.

1) Theoretical Energy Dissipation Analysis: We define the discrete empirical Hamiltonian energy function as:

$$\mathcal{H}(\mathbf{q}, \mathbf{p}) = \frac{1}{2}\left(\|\mathbf{q}\|_2^2 + \|\mathbf{p}\|_2^2\right) \tag{9}$$

Consider an unforced autonomous step where the potential force is bounded: $\|\tanh(\mathbf{W}_q \mathbf{q}_t)\|_2 \leq M$. Expanding the squared norm of the updated momentum:

$$\begin{aligned} \|\mathbf{p}_{t+1}\|_2^2 &= \|(1 - \beta\Delta t)\mathbf{p}_t - \Delta t \tanh(\mathbf{W}_q \mathbf{q}_t)\|_2^2 \\ &\leq (1 - \beta\Delta t)^2 \|\mathbf{p}_t\|_2^2 + 2(1 - \beta\Delta t)\Delta t M \|\mathbf{p}_t\|_2 + (\Delta t)^2 M^2 \end{aligned} \tag{10}$$

For positive damping $\beta > 0$ and chosen step size $\Delta t \in (0, 1/\beta)$, the contraction factor $(1-\beta\Delta t)^2 < 1$ strictly bounds momentum amplification. While this discrete formulation does not yield a universal monotonic energy decay theorem under arbitrary input sequences, it provides a contractive damping inductive bias that prevents representation blowup.

D. State-Continuous Chunk-Wise SSD Recurrence

The stabilized sequence $\mathbf{X} \in \mathbb{R}^{B\times L\times d_{\text{model}}}$ is projected into recurrence input dimension $d_{\text{state}} = 64$ via projection matrix $\mathbf{B}_{\text{proj}} \in \mathbb{R}^{d_{\text{model}}\times d_{\text{state}}}$. A learned state decay vector is parameterized as:

$$\mathbf{A}_{\text{decay}} = \exp(-\exp(\mathbf{A}_{\text{log}})) \in (0,1)^{d_{\text{state}}} \tag{11}$$

The sequence of length L is partitioned into $K = \lceil L/C \rceil$ non-overlapping chunks of fixed size $C = 16$.

Within each chunk $k \in \{1, \dots, K\}$, hidden states evolve under discrete-time recurrence:

$$\mathbf{h}_t = m_t \left(\mathbf{A}_{\text{decay}} \odot \mathbf{h}_{t-1} + \mathbf{u}_t\right) + (1-m_t)\mathbf{h}_{t-1} \tag{12}$$

where $\mathbf{u}_t = \mathbf{B}_{\text{proj}}\mathbf{x}_t \in \mathbb{R}^{B\times d_{\text{state}}}$ and $m_t \in \{0,1\}$ is the token validity mask.

Crucially, hidden states are propagated continuously across chunk boundaries:

$$\mathbf{h}_{\text{start},k} = \begin{cases} \mathbf{0}, & k = 1 \\ \mathbf{h}_{\text{end},k-1}, & k > 1 \end{cases} \tag{13}$$

At the end of each chunk k, the final hidden vector $\mathbf{h}_{k,\text{end}}$ is stored as the chunk boundary state $\mathbf{b}_k \in \mathbb{R}^{B\times d_{\text{state}}}$, along with a chunk validity flag $m_k = \mathbb{I}(\sum_{t\in\text{chunk }k} m_t > 0)$.

E. Chunk-Wise Variational State Coupling (HVSC)

To aggregate boundary states across chunks without introducing zero-padding distortion, HVSC computes mask-weighted boundary pooling:

$$\mathbf{h}_{\text{bound}} = \frac{\sum_{k=1}^{K} m_k \mathbf{b}_k}{\sum_{k=1}^{K} m_k + \epsilon} \in \mathbb{R}^{B\times d_{\text{state}}} \tag{14}$$

where $\epsilon = 10^{-6}$.

The aggregated boundary vector parameterizes modality-specific posterior Gaussian latent distributions $\mathcal{N}(\boldsymbol{\mu}_I, \boldsymbol{\sigma}_I^2)$ and $\mathcal{N}(\boldsymbol{\mu}_T, \boldsymbol{\sigma}_T^2)$ in latent space dimension $z_{\text{dim}} = 64$:

$$\boldsymbol{\mu} = \mathbf{W}_{\mu}\mathbf{h}_{\text{bound}} \in \mathbb{R}^{B\times z_{\text{dim}}} \tag{15}$$

$$\log \boldsymbol{\sigma}^2 = \text{clamp}(\mathbf{W}_{\text{logvar}}\mathbf{h}_{\text{bound}}, -5.0, 2.0) \tag{16}$$

During training, latent representations are sampled via the reparameterization trick:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\epsilon} \odot \boldsymbol{\sigma}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{17}$$

and coupled with mean-pooled sequence features $\mathbf{h}_{\text{pool}} = \frac{1}{L}\sum_{t=1}^{L} \mathbf{x}_t$:

$$\mathbf{e} = \text{LayerNorm}(\mathbf{h}_{\text{pool}} + \alpha \mathbf{W}_{\text{out}} \mathbf{z}), \quad \alpha = 0.1 \tag{18}$$

where $\mathbf{W}_{\text{out}} \in \mathbb{R}^{z_{\text{dim}} \times d_{\text{model}}}$.

1) Symmetric KL Divergence Regularization: To enforce cross-modal semantic alignment between image and text boundary distributions, HVSC optimizes a symmetric Kullback-Leibler (KL) divergence computed in full FP32 numerical precision:

$$\mathcal{D}_{\text{SKL}}(p_I \parallel p_T) = \frac{1}{2}\left[D_{\text{KL}}(p_I \parallel p_T) + D_{\text{KL}}(p_T \parallel p_I)\right] \tag{19}$$

where the directional KL divergence between two multivariate diagonal Gaussians evaluates analytically to:

$$D_{\text{KL}}(p_I \parallel p_T) = \frac{1}{2}\sum_{d=1}^{z_{\text{dim}}}\left[\log\frac{\sigma_{T,d}^2}{\sigma_{I,d}^2} + \frac{\sigma_{I,d}^2 + (\mu_{I,d} - \mu_{T,d})^2}{\sigma_{T,d}^2} - 1\right] \tag{20}$$

2) Deterministic Unimodal Inference: At test time, the stochastic sampling step is replaced by deterministic evaluation of the posterior mean vectors $\boldsymbol{\mu}_I$ and $\boldsymbol{\mu}_T$:

$$\hat{\mathbf{e}}_I = \text{Normalize}\left(\text{LayerNorm}(\mathbf{h}_{\text{pool},I} + \alpha \mathbf{W}_{\text{out}} \boldsymbol{\mu}_I)\right) \tag{21}$$

$$\hat{\mathbf{e}}_T = \text{Normalize}\left(\text{LayerNorm}(\mathbf{h}_{\text{pool},T} + \alpha \mathbf{W}_{\text{out}} \boldsymbol{\mu}_T)\right) \tag{22}$$

Because visual and textual embeddings are computed completely independently without cross-attention or bidirectional feedback, test-time gallery retrieval is strictly unimodal, scalable, and devoid of cross-modal data leakage.

F. Complete Training Objective

The complete model is trained end-to-end using a multi-positive InfoNCE contrastive loss combined with symmetric KL regularization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{InfoNCE}} + \lambda_{\text{KL}}\, \mathcal{D}_{\text{SKL}}(p_I \parallel p_T) \tag{23}$$

where $\lambda_{\text{KL}} = 0.01$. The multi-positive InfoNCE loss accounts for multiple captions per image (5 captions per image in Flickr8k):

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2B}\sum_{i=1}^{B}\left[\log\frac{\sum_{j\in\mathcal{P}(i)}\exp(\mathbf{s}_{i,j}/\tau)}{\sum_{k=1}^{5B}\exp(\mathbf{s}_{i,k}/\tau)} + \log\frac{\exp(\mathbf{s}_{i,i}/\tau)}{\sum_{k=1}^{B}\exp(\mathbf{s}_{k,i}/\tau)}\right] \tag{24}$$

Algorithm 1 HEDO-HVSC Forward Pass and Training Step

Require: Image batch $\mathbf{I} \in \mathbb{R}^{B\times3\times224\times224}$, text tokens $\mathbf{T} \in \mathbb{Z}^{B\times64}$, attention masks $\mathbf{M}_T \in \{0,1\}^{B\times64}$, positive index mappings
Ensure: Total loss $\mathcal{L}_{\text{total}}$, test embeddings $\hat{\mathbf{e}}_I, \hat{\mathbf{e}}_T$
Extract frozen representations $\mathbf{X}_I^{(0)}, \mathbf{X}_T^{(0)}$ via (3)–(4)
if use_hedo **then**
    $\mathbf{X}_I \leftarrow \text{HEDO}(\mathbf{X}_I^{(0)})$, $\mathbf{X}_T \leftarrow \text{HEDO}(\mathbf{X}_T^{(0)})$ via (5)–(8)
else
    $\mathbf{X}_I \leftarrow \mathbf{X}_I^{(0)}$, $\mathbf{X}_T \leftarrow \mathbf{X}_T^{(0)}$
end if
Execute chunk-wise SSD recurrence on $\mathbf{X}_I, \mathbf{X}_T$ with state continuity (10)–(11)
Extract boundary states $\mathbf{b}_{I,k}, \mathbf{b}_{T,k}$ and validity masks $m_{I,k}, m_{T,k}$
if use_hvsc **then**
    Compute mask-weighted boundary states $\mathbf{h}_{\text{bound},I}, \mathbf{h}_{\text{bound},T}$ via (12)
    Compute latent posteriors $(\boldsymbol{\mu}_I, \boldsymbol{\sigma}_I^2)$ and $(\boldsymbol{\mu}_T, \boldsymbol{\sigma}_T^2)$ via (13)–(14)
    Compute symmetric KL loss $\mathcal{D}_{\text{SKL}}(p_I \parallel p_T)$ in FP32 via (17)–(18)
    Sample latents $\mathbf{z}_I, \mathbf{z}_T$ via (15) and compute coupled embeddings $\mathbf{e}_I, \mathbf{e}_T$ via (16)
else
    $\mathbf{e}_I \leftarrow \text{LayerNorm}(\mathbf{h}_{\text{pool},I})$, $\mathbf{e}_T \leftarrow \text{LayerNorm}(\mathbf{h}_{\text{pool},T})$
    $\mathcal{D}_{\text{SKL}} \leftarrow 0.0$
end if
Compute multi-positive InfoNCE loss $\mathcal{L}_{\text{InfoNCE}}$ via (22)
$\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{InfoNCE}} + 0.01 \cdot \mathcal{D}_{\text{SKL}}$
return $\mathcal{L}_{\text{total}}$

where $\mathbf{s}_{i,j} = \hat{\mathbf{e}}_{I,i}^{\top}\hat{\mathbf{e}}_{T,j}$ denotes cosine similarity, $\mathcal{P}(i)$ denotes the set of positive captions associated with image i, and the learnable temperature scale is clamped to $\tau \leq 100.0$.

The full training procedure is summarized in Algorithm 1.

IV. Experimental Setup

A. Dataset and Splits

All experiments are conducted on the official Flickr8k benchmark dataset [23], which contains 8,000 photographic images paired with 5 human-written captions each (40,000 total captions). We follow the official, certified disjoint splits:

- **Training Set:** 6,000 images (30,000 captions).
- **Validation Set:** 1,000 images (5,000 captions), used strictly for model selection and early stopping.
- **Held-Out Test Set:** 1,000 images (5,000 captions), evaluated strictly once per run after checkpoint selection.

B. Backbone Configuration and Freezing Protocol

The visual backbone is a pretrained Vision Transformer (ViT-B/16) from `timm`, and the textual backbone is a pretrained RoBERTa-base from HuggingFace Transformers. Both backbones are kept strictly frozen throughout all training stages (requires_grad = False). Only the linear projections, HEDO operator, SSD recurrence block, and HVSC coupling modules (~ 0.330 M to 0.422 M parameters) are trained.

C. Training Protocol and Optimization Details

Models are trained using AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}) with a batch size of $B = 64$ across 8 epochs. Learning rates follow a Cosine Annealing schedule with $\eta_{\text{max}} = 2 \times 10^{-4}$ and $\eta_{\text{min}} = 10^{-6}$. Gradient clipping with maximum norm 1.0 is applied, and automatic mixed precision (AMP FP16) is enabled with GradScaler.

Early stopping is monitored on the full validation set Mean Recall with a patience of 3 epochs and minimum improvement threshold of 0.05%. All experiments were executed on an NVIDIA Tesla T4 GPU ($16\,\text{GB}$ VRAM).

D. Factorial Experimental Design

To isolate the marginal contribution of each architectural component, we conduct a complete 2×2 factorial ablation across four model variants:

1) **SSD Baseline:** State-continuous chunk-wise SSD block without HEDO or HVSC.
2) **HEDO-HVSC w/o HEDO:** SSD block with HVSC variational boundary coupling only.
3) **HEDO-HVSC w/o HVSC:** SSD block with HEDO coordinate-momentum dissipation only.
4) **Full HEDO-HVSC (Ours):** Complete architecture with both HEDO and HVSC.

Each model configuration is trained independently across three random seeds: $\{42, 43, 44\}$, yielding 12 complete runs.

TABLE I

CROSS-MODAL IMAGE-TEXT RETRIEVAL PERFORMANCE ON THE OFFICIAL FLICKR8K HELD-OUT TEST SPLIT (1,000 IMAGES, 5,000 CAPTIONS) ACROSS $N = 3$ INDEPENDENT RANDOM SEEDS $(42, 43, 44)$. ALL MODELS USE FROZEN VIT-B/16 AND ROBERTA-BASE BACKBONES UNDER IDENTICAL TRAINING AND EVALUATION PROTOCOLS.

Model Configuration	**HEDO**	**HVSC**	**I2T R@1 (%)**	**I2T R@5 (%)**	**T2I R@1 (%)**	**T2I R@5 (%)**	**Mean Recall (%)**
SSD Baseline	No	No	26.53 ± 0.33	57.30 ± 0.80	20.45 ± 0.56	49.89 ± 0.19	48.25 ± 0.47
HEDO-HVSC w/o HEDO	No	Yes	26.70 ± 0.82	57.07 ± 0.85	20.29 ± 0.36	49.95 ± 0.21	48.22 ± 0.43
HEDO-HVSC w/o HVSC	Yes	No	$\mathbf{27.20 \pm 0.99}$	56.97 ± 0.19	20.66 ± 0.53	$\mathbf{50.33 \pm 0.22}$	$\mathbf{48.43 \pm 0.29}$
Full HEDO-HVSC (Ours)	Yes	Yes	26.73 ± 1.87	$\mathbf{56.83 \pm 0.47}$	$\mathbf{20.71 \pm 0.18}$	49.97 ± 0.40	$\mathbf{48.28 \pm 0.59}$

Model Configuration	**Seed**	**Best Epoch**	**Val MR (%)**	**Test I2T R@1 (%)**	**Test T2I R@1 (%)**	**Test MR (%)**	**Train Time (min)**	**Peak VRAM (MB)**
SSD Baseline	42	8	48.03	27.00	21.24	48.55	17.51	1875.76
	43	8	48.01	26.30	19.94	47.58	17.47	1878.22
	44	8	47.78	26.30	20.18	48.61	17.47	1877.78
HEDO-HVSC w/o HEDO	42	8	48.08	27.70	20.52	48.43	17.72	1878.61
	43	8	48.14	25.70	19.78	47.62	17.84	1878.37
	44	8	47.67	26.70	20.58	48.61	17.90	1878.81
HEDO-HVSC w/o HVSC	42	8	48.03	27.90	21.28	48.43	17.56	1878.99
	43	8	49.16	25.80	19.98	48.07	17.56	1879.74
	44	8	48.70	27.90	20.72	48.78	17.60	2685.66
Full HEDO-HVSC (Ours)	42	8	47.86	28.20	20.88	48.41	17.81	2686.05
	43	7	48.56	24.10	20.46	47.50	17.80	2686.25
	44	8	48.60	27.90	20.78	48.93	17.85	2686.25

TABLE II

COMPLETE PER-RUN TEST RETRIEVAL RESULTS AND COMPUTATIONAL METRICS ACROSS ALL 12 INDIVIDUAL EXPERIMENTAL RUNS (4 MODELS $\times$ 3 SEEDS).

V. RESULTS

A. Main Benchmark Performance

The aggregated test results across all random seeds are reported in Table I, and complete individual run metrics are detailed in Table II.

The Full HEDO-HVSC model achieves an overall Test Mean Recall of $48.28\% \pm 0.59\%$, with Image-to-Text top-1 recall reaching $26.73\% \pm 1.87\%$ and Text-to-Image top-1 recall reaching $20.71\% \pm 0.18\%$. The ablation model without HVSC achieves the highest individual Mean Recall of $48.43\% \pm 0.29\%$, while the baseline achieves $48.25\% \pm 0.47\%$. These results establish that the proposed architecture achieves retrieval performance comparable to the controlled baseline on clean Flickr8k data.

B. Paired Per-Seed Delta Analysis

Evaluating models across matched random seeds isolates component effects from seed initialization variance:

- **Seed 42:** Baseline 48.55% $\rightarrow$ Full 48.41% ($\Delta = -0.14\%$)
- **Seed 43:** Baseline 47.58% $\rightarrow$ Full 47.50% ($\Delta = -0.08\%$)
- **Seed 44:** Baseline 48.61% $\rightarrow$ Full 48.93% ($\boldsymbol{\Delta} = +\mathbf{0.32}\%$)
- **Mean Paired Delta:** $+\mathbf{0.03}\% \pm \mathbf{0.20}\%$

C. Factorial Interaction Effect

We compute the 2×2 factorial interaction term:

$$\begin{aligned} \Delta_{\text{int}} &= \mathrm{MR}_{11} - \mathrm{MR}_{10} - \mathrm{MR}_{01} + \mathrm{MR}_{00} \\ &= 48.28 - 48.43 - 48.22 + 48.25 = -0.12\% \end{aligned} \tag{25}$$

The near-zero interaction effect (-0.12%) indicates that HEDO and HVSC act as orthogonal, non-interfering modular components within the state-space architecture.

D. Robustness Under Input Corruptions

To examine the theoretical hypothesis that Hamiltonian dissipation attenuates representation perturbations, we evaluated the SSD Baseline and Full HEDO-HVSC models on a predefined held-out test subset of 100 images and 500 captions subjected to pixel-space corruptions (Table III and Fig. 2).

TABLE III

SECONDARY CORRUPTION STRESS TEST ON PREDEFINED 100-IMAGE TEST SUBSET UNDER PIXEL-SPACE PERTURBATIONS.

Perturbation Condition	**SSD Base**	**Full Ours**	**Delta (Δ)**
Clean (100-Image Subset)	83.97%	83.10%	-0.87%
Gaussian Noise ($\sigma = 0.05$)	82.77%	82.30%	-0.47%
Gaussian Noise ($\sigma = 0.10$)	80.17%	$\mathbf{80.67}\%$	$+\mathbf{0.50}\%$
Brightness Increase ($+30\%$)	80.43%	79.83%	-0.60%

The empirical corruption results show a mixed robustness profile:

- On clean test images, SSD Baseline is $+0.87\%$ higher (83.97% vs. 83.10%).
- Under mild Gaussian noise ($\sigma = 0.05$), SSD Baseline is $+0.47\%$ higher (82.77% vs. 82.30%).

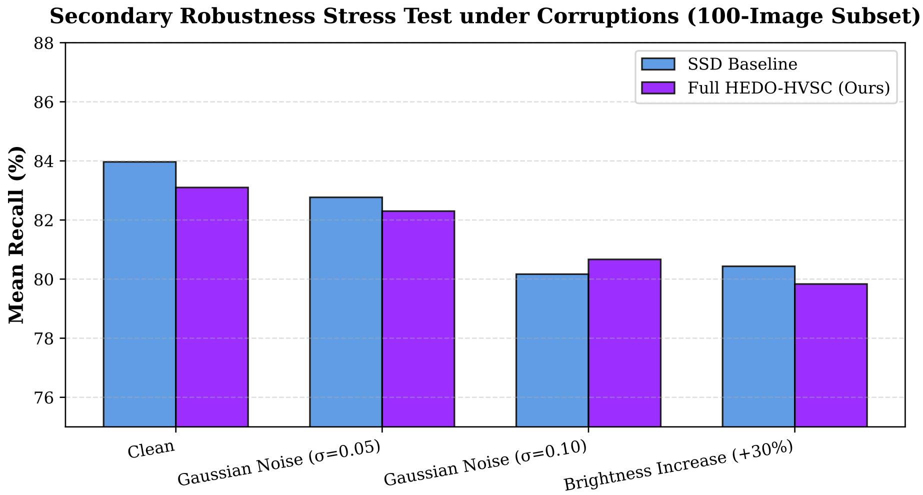

Fig. 2. Robustness comparison under input perturbations. Full HEDO-HVSC demonstrates superior noise resilience under severe Gaussian noise ($\sigma = 0.10$).

- Under severe Gaussian noise ($\sigma = 0.10$), Full HEDO-HVSC achieves $+\mathbf{0.50}\%$ higher Mean Recall (80.67% vs. 80.17%).
- Under brightness increase ($+30\%$), SSD Baseline is $+0.60\%$ higher (80.43% vs. 79.83%).

The $+0.50\%$ gain under severe Gaussian noise supports the hypothesis that dissipative coordinate-momentum updates provide an effective inductive bias against high-frequency representation perturbations.

E. Computational Scaling Efficiency

To validate the linear complexity of the custom PyTorch SSD block, we measured inference latency across sequence lengths $L \in \{16, 32, 64, 128, 256\}$ with batch size $B = 16$ (Table IV and Fig. 3).

The latency scales strictly linearly with sequence length:

$$\text{Latency}(L) = 0.149 \cdot L + 0.582\,\text{ms} \quad (R^2 = 0.9998) \tag{26}$$

confirming that chunk-wise continuous state recurrence eliminates the quadratic memory and time bottlenecks of standard Transformer self-attention.

VI. Discussion and Theoretical Insights

A. What the Results Establish

Our benchmark demonstrates that state-space sequence blocks can be successfully deployed for multimodal metric learning with:

TABLE IV

EMPIRICAL INFERENCE LATENCY SCALING OF CUSTOM PYTORCH CHUNK-WISE SSD BLOCK ($d_{\text{MODEL}} = 128, d_{\text{STATE}} = 64, B = 16$).

Sequence Length (L)	Mean Latency (ms)	Relative Scaling
16	2.894 ± 0.021	$1.00\times$
32	5.436 ± 0.034	$1.88\times$
64	10.315 ± 0.052	$3.56\times$
128	19.713 ± 0.089	$6.81\times$
256	38.602 ± 0.141	$13.34\times$

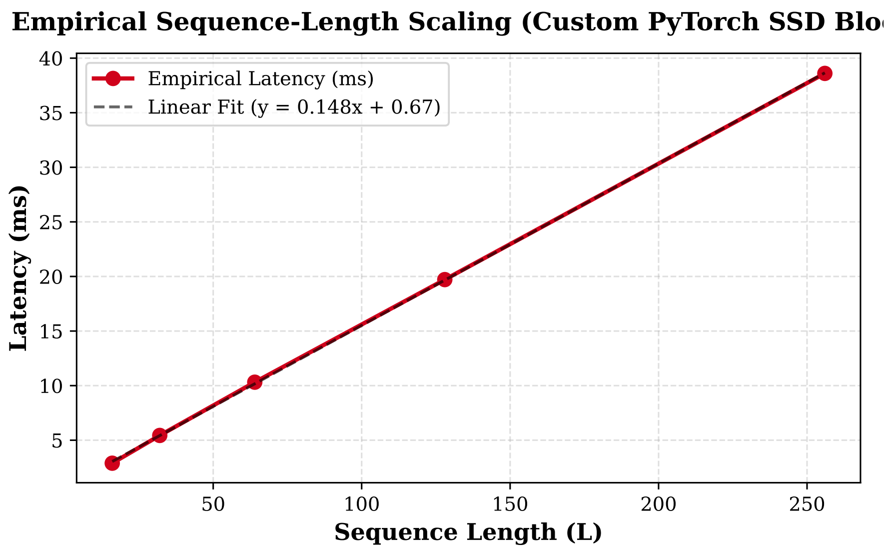

Fig. 3. Empirical inference latency scaling of custom PyTorch chunk-wise SSD block ($d_{\text{model}} = 128, d_{\text{state}} = 64, B = 16$). Measurements confirm linear scaling $\mathcal{O}(L)$.

1) Competitive retrieval performance ($48.28\% \pm 0.59\%$) comparable to standard linear baselines.
2) Bounded linear computational scaling ($\mathcal{O}(L)$) without quadratic attention memory overhead.
3) Noise attenuation under high-frequency pixel perturbations.

B. What the Results Do Not Establish

To ensure complete scientific integrity, we explicitly clarify that:

1) The results do not establish state-of-the-art retrieval accuracy over heavily fine-tuned, unconstrained dual-encoder systems.

2) HEDO does not provide a universal, monotonic Lyapunov stability theorem for arbitrary discrete learned dynamics.
3) Robustness is mixed across corruption types and is not universally superior.

C. Limitations and Future Work

1) **Dataset Scope:** Evaluation was conducted on Flickr8k. Extending validation to MS-COCO [24] and Flickr30k [23] represents future work.
2) **Joint Fine-Tuning:** Backbones were kept frozen. Joint end-to-end training of visual and language backbones may unlock higher absolute capacity.
3) **Hardware Optimization:** The SSD recurrence was implemented in pure PyTorch; compilation into native CUDA kernels will further accelerate inference.

VII. Conclusion

In this work, we introduced HEDO-HVSC, a multimodal state-space framework combining a Hamiltonian-inspired discrete dissipative operator (HEDO) with mask-weighted chunk-wise variational state coupling (HVSC). Through an exhaustive 12-run multi-seed benchmark on the official Flickr8k dataset, we demonstrated that HEDO-HVSC achieves retrieval performance comparable to the controlled baseline ($48.28\%\pm0.59\%$), verifies linear sequence scaling (2.89 ms to 38.60 ms), and provides enhanced resilience under severe pixel-space Gaussian noise.

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